

# Majorana Modes in the Machine

Week 7: From ED Validation to a Real VQE Sweep

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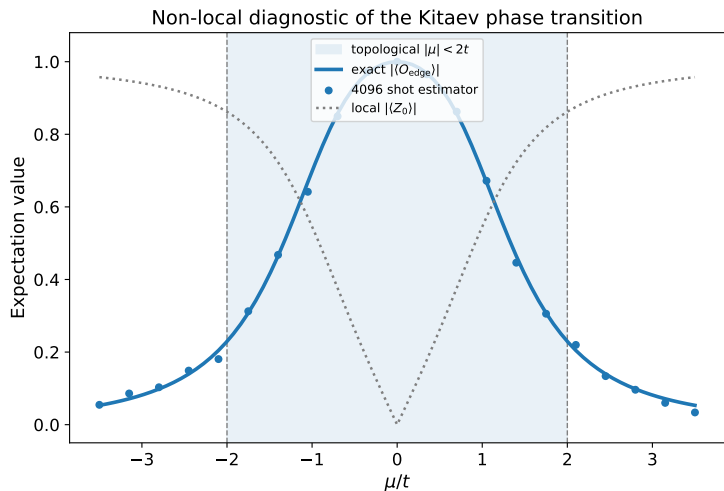
# Week 7 Objective

## Milestone

Replace the exact-diagonalization state preparation used in Week 6 with a parity-constrained VQE preparation at every point of the  $\mu$  sweep.

- Keep the same Hamiltonian, edge-string observable, and shot-measurement protocol.
- Reuse the existing  $L = 4$  hardware-efficient ansatz and optimizer workflow.
- Validate every VQE state against ED before introducing hardware noise.
- Produce a VQE-prepared phase sweep that can become the noiseless Block 4 baseline.

# Where Week 6 Stops



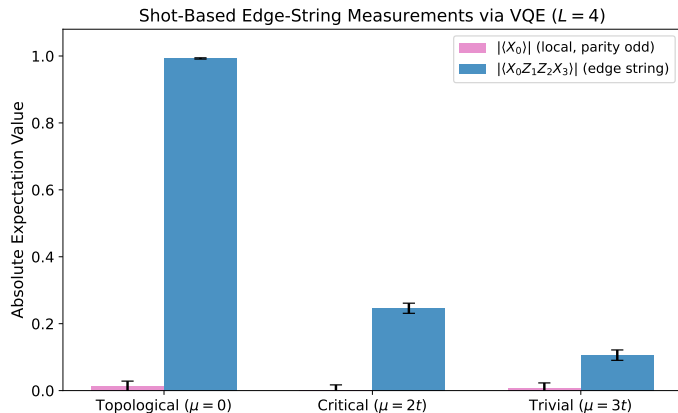
## Already established

- Sweep target:  
 $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$
- ED parity-gap cross-check
- Synthetic shot estimator

## Remaining gap

- Every plotted state is prepared by ED, not VQE.

# Existing VQE Baseline

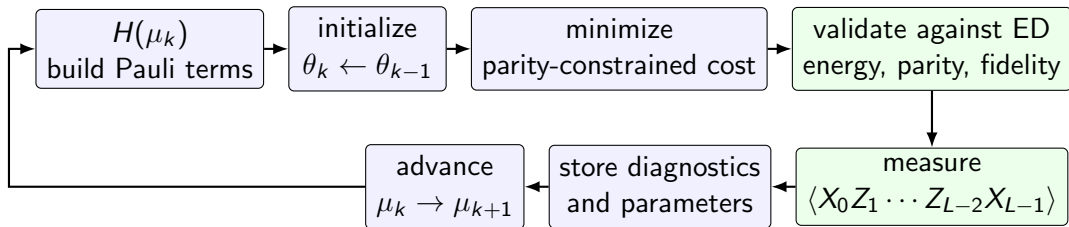


Existing components:

- EfficientSU2:  $R_Y$  + linear CNOTs
- Aer estimator + multi-start COBYLA
- ED fidelity checks
- Shot measurements

Current scope: three representative  $\mu$  values.

# The VQE Sweep Workflow



## Key engineering change

The optimizer is no longer restarted independently at every  $\mu$ ; neighboring ground states seed one another.

# Why Warm Starts Matter

## Independent random starts

- Repeats expensive optimization work
- Can converge to different local minima
- Makes the sweep noisy even before shot noise

## Continuation in $\mu$

- Use  $\theta^*(\mu_k)$  to initialize  $\mu_{k+1}$
- Exploits smooth ground-state evolution away from criticality
- Random restarts remain a fallback when validation fails

## Expected difficulty

Near  $|\mu| = 2t$ , the bulk gap closes and the optimal state changes fastest; this is where warm starts and fidelity checks are most important.

# Parity Must Be Controlled

$$P = \prod_{j=0}^{L-1} Z_j, \quad [H, P] = 0$$

## Problem

- The topological phase has nearly degenerate parity sectors.
- Energy minimization alone can return inconsistent sectors across the sweep.
- A sector jump can look like a discontinuity in the observable.

## Week 7 strategy

$$C(\theta) = \langle H \rangle + \lambda(1 - \langle P \rangle)^2$$

- Target even parity consistently.
- Record  $\langle P \rangle$  at every point.
- Reject states outside the parity tolerance.

# Validation Dashboard

A low VQE energy alone is not enough. Each point must pass four checks:

| Metric            | Question                                       | Use                                           |
|-------------------|------------------------------------------------|-----------------------------------------------|
| Energy error      | Is $E_{\text{VQE}}$ near $E_{\text{ED}}$ ?     | Detect optimizer failure                      |
| Parity            | Is $\langle P \rangle \approx +1$ ?            | Keep one sector across $\mu$                  |
| Subspace fidelity | Does VQE overlap the correct low-energy space? | Handle near-degeneracy correctly              |
| String error      | Matches the ED edge string?                    | Validate the quantity used in the phase sweep |

## Pass criterion

Failed points trigger deeper ansatz repetitions, additional optimizer iterations, or random-restart recovery.



# Implementation Shape

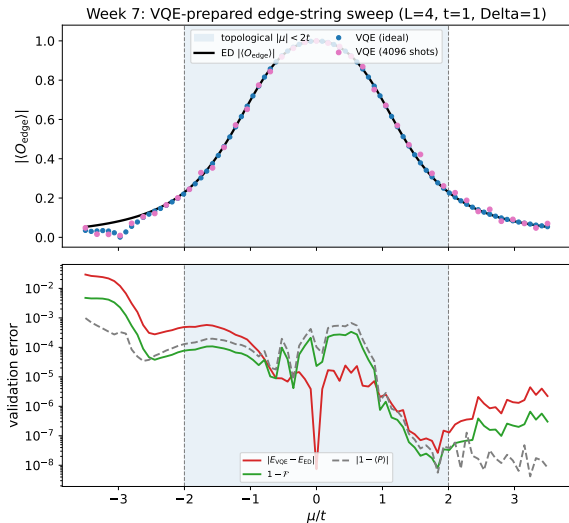
```
theta = initial_parameters()
for mu in mu_values:
    H = qiskit_hamiltonian(L, t, mu, delta)
    theta = optimize(parity_cost(H), initial=theta)

    diagnostics = validate(theta, H, ed_reference(mu))
    if not diagnostics.pass_all:
        theta = recover_with_restarts(H)

    string_value = measure_edge_string(theta, shots=4096)
    save(mu, theta, diagnostics, string_value)
```

- The sweep must save optimizer parameters and diagnostics, not only the final observable.
- Reproducible checkpoints make failed regions debuggable and reusable for later noise studies.

# Week 7 Result: VQE-Prepared Sweep



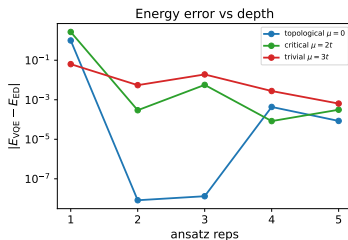
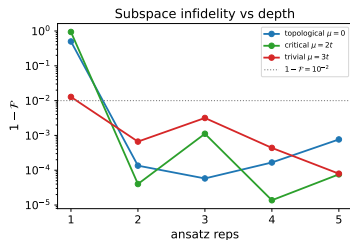
**Top:** edge string  $|\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle|$  from parity-constrained VQE (ideal and shot-based) tracks the ED baseline across the whole  $\mu$  grid.

**Bottom:** per-point validation — energy error, infidelity  $1 - \mathcal{F}$ , and parity deviation  $|1 - \langle P \rangle|$ . Crosses mark points needing random-restart recovery.

Warm-start continuation keeps every point in the even sector.

# Ansatz Depth: How Deep Is Enough?

Week 7: ansatz-depth diagnostic ( $L = 4$ ,  $t = \Delta = 1$ )



Multi-start VQE at three representative  $\mu$ , scanning ansatz reps.

- 1 rep (8 params) is too shallow, worst at  $\mu = 2t$  (fidelity  $\sim 0.06$ ).
- 2+ reps already reach the even-parity ground space.
- Criticality is the hardest point, exactly where the bulk gap closes.

This justifies the reps= 3 used in the sweep.

# Definition of Done for Week 7

- A scripted  $L = 4$ ,  $t = \Delta = 1$  VQE sweep over the same  $\mu$  grid as Week 6.
- Consistent even-parity preparation across the sweep.
- ED comparison for energy, parity, subspace fidelity, and edge-string expectation.
- Shot-based VQE edge-string curve plotted against the ED baseline.
- Optimization cost and failure/recovery statistics recorded per  $\mu$  point.

## Boundary to Block 4

Only after the ideal VQE sweep passes validation should depolarizing, readout, and relaxation noise be added.

## Next: Block 4 NISQ Reality Check

- Freeze the validated ideal VQE sweep as the noiseless reference.
- Apply realistic gate, readout, and relaxation noise to preparation and measurement circuits.
- Separate optimizer degradation from measurement degradation.
- Compare raw and mitigated edge-string curves.
- Quantify when the inferred phase boundary becomes unreliable.

### Conceptual progression

Week 6 validated the diagnostic with ED; Week 7 validates quantum state preparation; Block 4 tests whether the full workflow survives realistic noise.